

AI-Based Legal Document Analysis and Verdict Prediction using Machine Learning

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Abstract—Modern legal frameworks directly affect almost every aspect of our daily life, yet navigating the court system remains quite confusing for the average person. Ordinary citizens often face heavy financial losses and delays simply because they don't have clear insights into court proceedings, precedents, or how a lawsuit might turn out. To solve this issue, our project focuses on creating a machine learning framework that predicts court case outcomes based on historic legal records. Using basic NLP and text vectorization techniques, we extract important features from unstructured legal summaries and convert them into structured inputs. We evaluated five standard classification models Logistic Regression, DistilBERT, Multinomial Naive Bayes, LegalBERT-small, and Linear Support Vector Machines (SVM) using metrics like accuracy, precision, recall, and F1-score. From the experimental results, TF-IDF combined with Logistic Regression gave the highest precision and reliability among all tested algorithms. Overall, this system can work as a practical tool for legal professionals and everyday people to evaluate litigation risks before spending too much time and money.

Index Terms—Artificial Intelligence, Machine learning, Judicial Case, Outcome, Prediction, Civilians, performance comparison.

I. INTRODUCTION

Legal systems are essential for any organized society because they govern daily interactions, business activities, and public affairs. But as laws grow more complex every day, a big gap has formed between court systems and normal citizens. Since legal procedures are rarely taught in basic school curriculum, most people enter legal battles with very little understanding of the procedural steps, realistic timelines, or total costs involved. Because of this, litigants frequently end up taking wrong risks, getting stuck in long legal battles, and paying excessive legal fees.

When someone faces a potential court case, they have to deal with many uncertain decisions: finding the right lawyer, understanding court jurisdiction, estimating how long the case will run, and figuring out if they can afford it financially. The core problem is not just lack of information, but how hard it is to manually read through thousands of past rulings to see how similar cases were decided.

In recent years, machine learning and Natural Language Processing (NLP) has transformed how we handle large text

data. While traditional legal research depends fully on human experts to analyze past precedents, machine intelligence can easily process large amounts of text to spot hidden patterns. When applied to legal domains, predictive analytics can turn raw case summaries into practical outcome predictions.

In this paper, we explore computational methods for forecasting court decisions using historical legal records from Indian courts. We cleaned the unstructured text by removing unnecessary stop-words and converted case descriptions into numerical vectors for classification. We tested five popular models Logistic Regression, Naive Bayes, DistilBERT, LegalBERT-small and SVM to figure out which algorithm works best for legal text. Ultimately, this research aims to build a practical decision-support tool that brings more clarity to legal outcomes and helps both lawyers and ordinary people make better choices.

II. PREDICTIVE MODELS

The system we are designing is meant to predict the result of legal cases in India. It uses the case description along with other details such as the type of case and its category. The main goal is to classify each case into one of two groups: Accepted or Rejected. To choose the best model, we trained and tested several machine learning and transformer-based models using the same data set split to avoid data leakage. We then compared their performance using the validation results.

A. Multinomial Naive Bayes

Multinomial Naive Bayes is a machine learning method often used for text classification. It works well with text data and is easy to train and quick to run. In this project, we converted the legal case descriptions into numerical features using TF-IDF. The model then used these features to calculate the probability of a case being in the Accepted or Rejected group. The outcome with the higher probability was taken as the prediction.

B. Logistic Regression

Logistic Regression is a type of machine learning used for classification. In our system, it uses TF-IDF features from the case descriptions along with encoded case category and type. Since this is a binary task with two possible outcomes,

the model calculates the probability of both Accepted and Rejected. The class with the higher probability is chosen as the final prediction. We also balanced the classes during training to reduce any bias caused by unequal numbers of Accepted and Rejected cases.

Among all the models tested, Logistic Regression performed best. So, it was chosen as the final model.

C. Linear Support Vector Machine

Linear Support Vector Machine (SVM) is a supervised learning method that finds a line or boundary to separate different classes. Its goal is to find the best separation while keeping the distance between the classes as wide as possible.

In our study, Linear SVM was trained using TF-IDF features from case descriptions along with other case-related metadata. It was included to compare a margin-based approach with the probability-based Logistic Regression model.

D. DistilBERT

DistilBERT is a lightweight version of the BERT model, known for its efficiency. Unlike traditional text-based methods like TF-IDF, DistilBERT can understand the context and sequence of words in a sentence. We fine-tuned DistilBERT on the legal case dataset to predict whether a case is Accepted or Rejected. Its performance was compared with traditional models to see if a transformer-based approach would be more effective for this task.

E. LegalBERT-small

LegalBERT-small is a transformer model specifically trained on legal language. Because of this, it can better understand the context and specialized terms used in legal documents, which might not be captured well by simple word frequency methods.

In the system, LegalBERT-small was fine-tuned on the legal case dataset. To address the slight imbalance between Accepted and Rejected cases, we used a weighted loss function during training. Its performance was then compared with Logistic Regression, Linear SVM, Multinomial Naive Bayes, and DistilBERT.

F. Selected Final Model

After comparing the validation results of all the models, TF-IDF combined with Logistic Regression was selected as the final model because it achieved the highest Macro-F1 score.

This model was then retrained using both the training and validation data. After that, it was only tested once on data it hadn't seen before.

The final results were:

- * Test Accuracy: 61.06%
- * Macro-F1 Score: 60.99%
- * Number of Test Cases: 5,662

The results indicate that the descriptions and metadata of legal cases can reveal useful patterns that can help in research-related predictions.

III. DATASET AND FEATURES

For this study, a dataset that includes records of Indian legal cases was used to create a system that can predict how legal cases will turn out. The original dataset had 53,446 rows. After cleaning the data, removing any incomplete records, and getting rid of duplicate cases, 44,655 unique cases were left for the three-class experiment.

For the main task of binary classification, only the cases marked as Accepted and Rejected were included. The cases that were in the Other category were removed so the model could focus on a clear two-class prediction problem. After this filtering, the final binary dataset included 37,747 unique legal cases.

The binary dataset was split into training, validation, and test sets using a method that ensures the data is split fairly and without leaking information.

The distribution of the dataset is as follows:

TABLE I
DATASET DISTRIBUTION

Dataset	Number of Cases	Percentage
Training Set	26,423	70%
Validation Set	5,662	15%
Test Set	5,662	15%
Total	37,747	100%

The splitting was done at the unique case-ID level. This ensured that any particular case would not appear in more than one split. The test set was kept separate throughout the model development and was only used for the final evaluation.

The following features were used as inputs for the prediction model:

- case_info: Text containing the description and important facts of the legal case
- case_category: General category of the case, such as civil or criminal
- case_type: More specific type of case, such as appeal, writ petition, or special leave petition

The 'case_info' feature contains text, so it was converted into numerical features using TF-IDF vectorization. The categorical features 'case_category' and 'case_type' were converted into numerical form using one-hot encoding.

The target variable used for the binary prediction was 'label', which has two possible outcomes: Accepted and Rejected.

Special care was taken to prevent data leakage during the dataset preparation. Features such as 'judgement' and 'proof_sentence' were not used because they could directly reveal the final outcome of a case. Similarly, 'tokens' and 'sentences' were excluded because they might have been generated from texts that already contained information related to the case outcome.

IV. PERFORMANCE METRICS

To compare the performance of different predictive models, several commonly used classification metrics were considered.

These included Accuracy, Precision, Recall, F1-score, and Macro-F1 score. The values of these metrics were calculated using the confusion matrix. For the binary classification problem, the confusion matrix can be represented as follows:

TABLE II
CONFUSION MATRIX TERMS

Actual/Predicted	Predicted-Accepted case	Predicted-Rejected case
Actual Accepted	True Positive (TP)	False Negative (FN)
Actual Rejected	False Positive (FP)	True Negative (TN)

A. Accuracy

Accuracy shows the overall percentage of predictions made correctly by the model. It considers both the correctly predicted Accepted and Rejected cases.

$$Accuracy \Rightarrow \frac{TP + TN}{TP + TN + FP + FN}$$

B. Precision

Precision tells us how many of the cases predicted as a particular class were actually members of that class.

$$Precision \Rightarrow \frac{TP}{TP + FP}$$

C. Recall

Recall measures how many of the actual cases belonging to a particular class were correctly identified by the model.

$$Recall \Rightarrow \frac{TP}{TP + FN}$$

D. F1-Score

The F1-score combines Precision and Recall into a single value. It is calculated as the harmonic mean of the two measures.

$$F1 \Rightarrow 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

E. Macro-F1 Score

Macro-F1 calculates the F1-score separately for the Accepted and Rejected classes and then takes the average of the two scores.

$$Macro-F1 \Rightarrow \frac{F1_{Accepted} + F1_{Rejected}}{2}$$

In this study, Macro-F1 was considered the main metric for selecting the final model because it gives equal importance to both classes instead of allowing the performance of one class to dominate the overall score.

The selected TF-IDF with Logistic Regression model was finally evaluated on the unseen test set. The obtained results are as follows:

The results show that the proposed model is able to learn some useful patterns from the legal case descriptions and related case information. These patterns can be used for research-oriented prediction of legal case outcomes.

Final Model Performance on Unseen Test Data

Metric	Result
Test cases	5,662
Accuracy	61.06%
Macro-F1 score	60.99%
Accepted-class F1-score	59.41%
Rejected-class F1-score	62.57%

Fig. 1. Obtained test result

V. PARAMETERS

The different models used in this study were trained with specific settings to achieve their predictions. For the text-based models, the legal case descriptions were first turned into numerical features using TF-IDF vectorization. The maximum number of features was set to 100,000. Both individual words and pairs of words were included by setting the n-gram range to (1, 2). Terms that appeared in less than two case documents were excluded from the feature set.

For Logistic Regression, the SAGA solver was used. The regularization parameter was set to $C = 1.0$, and L2 regularization was applied. The maximum number of iterations was set to 1,000. During training, class weights were balanced to ensure both the Accepted and Rejected classes were given appropriate attention.

For the Support Vector Machine, a linear classifier was used. The model was trained using the TF-IDF features from the case descriptions along with the encoded values for case category and case type.

For Multinomial Naive Bayes, the standard Multinomial Naive Bayes classifier was used. This model was selected because it is commonly used for text classification and works well with the non-negative feature values generated by TF-IDF.

Two transformer-based models were also included for comparison. DistilBERT was trained for 3 epochs with a learning rate of 0.00002, a batch size of 4, and a maximum input length of 256 tokens. LegalBERT-small was trained for 4 epochs using the same learning rate of 0.00002 and a batch size of 4, while its maximum input length was 384 tokens. Weighted loss was used for LegalBERT-small to minimize the impact of the slight imbalance between the two classes.

The full dataset was split into training, validation, and test sets in the ratio of 70%, 15%, and 15%, respectively. A random seed of 42 was used to ensure the data-splitting and training setup could be reproduced if needed.

VI. EXPERIMENTATION AND RESULTS

The system was tested to predict legal verdicts, categorizing each case as either Accepted or Rejected. The final dataset included 37,747 unique legal cases. These were split into training, validation, and test sets in a ratio of 70%, 15%, and

15%, respectively. The training set had 26,423 cases, while the validation and test sets each had 5,662 cases.

Before training the models, the text from the case_info field was converted into numerical features using TF-IDF vectorization. The case_category and case_type fields were also converted into numerical form using one-hot encoding. Fields related to the judgment were not used as input features because they might give away information about the final result and cause data leakage.

The first set of experiments used Multinomial Naive Bayes, Logistic Regression, and Linear Support Vector Machine. In addition, DistilBERT and LegalBERT-small were fine-tuned for comparison with transformer-based methods. The models were mainly evaluated using Accuracy and Macro-F1 score based on the validation data. The validation results are as follows:

Experimental Results for Binary Verdict Prediction

Model	Evaluation Set	Accuracy	Macro-F1
Multinomial Naive Bayes	Validation	57.51%	55.45%
Linear SVM	Validation	59.61%	59.39%
Logistic Regression	Validation	60.56%	60.40%
DistilBERT	Validation	58.92%	56.91%
LegalBERT-small	Validation	58.65%	58.13%
Final Logistic Regression	Unseen Test	61.06%	60.99%

Fig. 2. Experimental result

Based on the validation results, Logistic Regression had the highest Macro-F1 score of 60.40% among all the models tested. Therefore, the combination of TF-IDF and Logistic Regression was chosen as the final model.

After choosing the model, Logistic Regression was retrained using both the training and validation data. The final model was then tested on the test set, which had not been used during training or model selection.

On the 5,662 unseen test cases, the model achieved an accuracy of 61.06% and a Macro-F1 score of 60.99%.

The final results are:

Final Model Performance

Final Model	Test Cases	Accuracy (%)	Macro-F1 (%)	Leakage Check
TF-IDF + Logistic Regression	5,662	61.06	60.99	Passed

Confusion Matrix

	Predicted Accepted	Predicted Rejected
Actual Accepted	1614	1053
Actual Rejected	1152	1843

Fig. 3. Final result

These results show that the model was able to pick up meaningful patterns from the descriptions of legal cases and related information. Logistic Regression outperformed other models, including the transformer-based ones, on this dataset.

At the same time, an accuracy of 61.06% shows that the model still has some limitations and is not yet reliable enough to make real judicial decisions. Therefore, the proposed system should be used as a research-oriented decision-support tool. It could be helpful for analyzing legal data and studying patterns in outcomes, but it should not be used as a substitute for professional legal judgment or judicial decisions.

CONCLUSION

Artificial Intelligence has become a key part of many fields where there is a need to analyze a lot of data and extract useful information. The growth of machine learning and natural language processing has also opened up new opportunities to use computational methods in areas like law.

The legal system handles a lot of information in the form of case descriptions, case categories, case types, and past case records. Manually going through all this information can take a lot of time. The proposed system looks into how machine learning techniques can be used to study this data and predict the likely outcome of an Indian legal case as either Accepted or Rejected.

In this project, several machine learning and transformer-based models were tested, including Multinomial Naive Bayes, Logistic Regression, Linear SVM, DistilBERT, and LegalBERT-small. After comparing their performance, TF-IDF with Logistic Regression was chosen as the final model because it gave the best validation Macro-F1 score.

The final model was tested on a set of 5,662 cases that it had not seen before. It achieved an accuracy of 61.06% and a Macro-F1 score of 60.99%. These results suggest that the information in case descriptions and related data contains useful patterns that can support research in predicting legal outcomes.

The main goal of this system is to offer a computational method that helps in analyzing legal cases and provides an early indication of a possible outcome. It can help save time during the initial stages of case evaluation and can serve as a research and decision-support tool.

However, it's important to note that the predictions from the system should not be treated as a final legal decision. Legal outcomes depend on many things like the full facts of a case, relevant laws, evidence, arguments, and judicial reasoning. Therefore, this system is meant for academic and research purposes only and should not replace professional legal advice or real judicial decisions.

Overall, this project shows the potential of using machine learning and natural language processing to analyze Indian legal case data. With more and better data, improved features specific to the legal domain, and further improvements in the models, the system could be made even more effective in future work.